

Methodology Report: Crossref Citation Tallying & DOI Resolution

This document explains the programmatic methodology used in the SuperDARN plotting scripts to fetch, resolve, and aggregate citation counts using the open Crossref Metadata Index.

1. Registry Ingestion & Pre-Processing

The script first downloads the official SuperDARN Canada Publications Registry as a CSV database from their server:

- **Source URL:**

https://sdc-serv.usask.ca/website Updating_tools/publication_reports/articles_2SWA2C.csv

- **Parsing Fields:** Reads the columns: DOI, Title, First Author, and Publication Date.

- **DOI Classification:** Records with a valid DOI (starting with 10.) are mapped directly to their publication year. Records with missing, empty, or placeholder DOIs (such as *'Incomplete Data'*) are sent to the fallback bibliographical resolution queue. This is crucial for the 179 papers published between 1995 and 2001, which lack DOIs in the database.

2. DOI-Based Batch Querying (For Papers with DOIs)

For the papers containing pre-existing DOIs (primarily 2002–2025), querying them individually would be slow and trigger rate limits. To solve this, the script performs batch querying:

- **Batching:** DOIs are split into groups of 50.

- **Filtering:** Joined into a comma-separated filter string: doi:10.1029/abc,doi:10.1016/xyz,...

- **Query Endpoint:** The script calls the Crossref /works API requesting selected fields to minimize payload size:

```
GET https://api.crossref.org/works?filter={filter_string_url_encoded}&rows=50&select=DOI,is-referenced-by-count
```

- **Data Extraction:** Reads the is-referenced-by-count field (the lifetime citations received across indexed publishers) and adds it to the tally for that paper's publication year.

3. Bibliographic Fallback Resolution (For Papers without DOIs)

For the 386 registry papers lacking DOIs (including all early papers from 1995–2001), the script falls back to a bibliographical search on the Crossref index:

- **Search Query:** Combines the paper title and the first author's name: '{title} {author}'.

- **API Call:** Queries Crossref for the single most relevant search match:

```
GET https://api.crossref.org/works?query={query_string_url_encoded}&rows=1&select=DOI,is-referenced-by-count
```

- **Tallying:** Selects the top search result, extracts its is-referenced-by-count citation metric, and attributes it to the publication year of the original registry record.

4. Year-by-Year Aggregation

Once both batches and resolved searches are processed, citations are grouped by publication year:

- **Annual Citations:** Computes the sum of citations of all papers published in year y .
- **Cumulative Citations:** The cumulative total of citations representing the impact of all papers published up to year y is computed using the running sum of these citation totals over time. This is implemented using NumPy's cumulative sum: `np.cumsum(annual_citations)`.